

Bayes Network Independence

Semantics: d-Separation

Let X, Y, Z be disjoint sets of nodes in a Bayes network with directed graph G .

Independent($X, Y \mid Z$) if and “almost only” if

X is d-separated from Y given Z , denoted

$d\text{-sep}(X, Y \mid Z)$. Latter holds just when

every node x in X is d-separated from every node y in Y given Z .

D-separation Continued

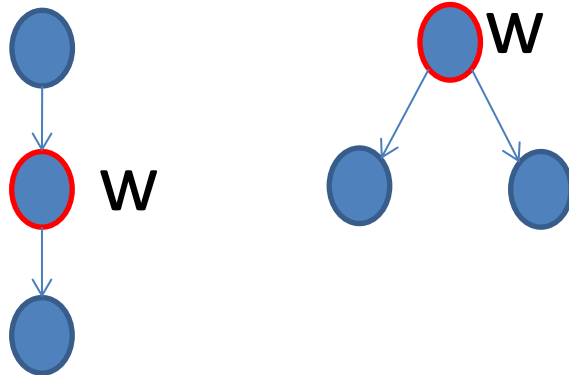
Node x is d-separated from node y given Z , denoted $d\text{-sep}(x,y|Z)$, just when every **undirected path from x to y is blocked by Z .**

An undirected path p from x to y is blocked by Z just when there exists a **node w on p that is blocked by Z .**

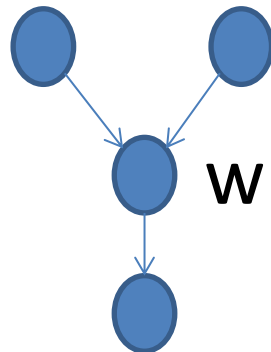
(w must be neither x , nor y , and the blocking is defined (below) relative to the path p .)

D-separation on path

Path is blocked by Z iff (passthrough or diverging and w in Z),

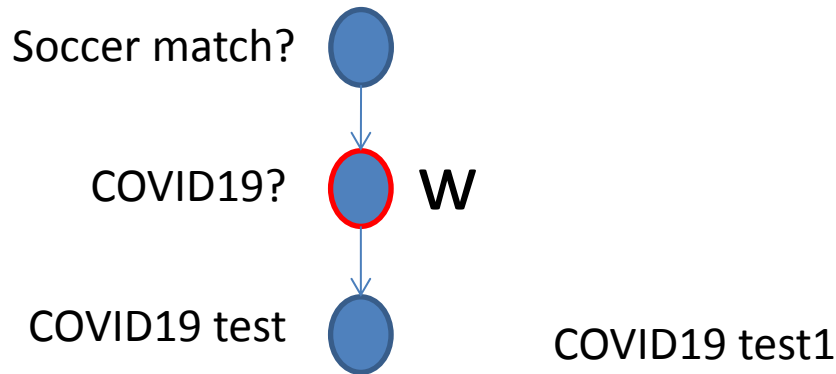


or (converging and neither w nor descendants are in Z).

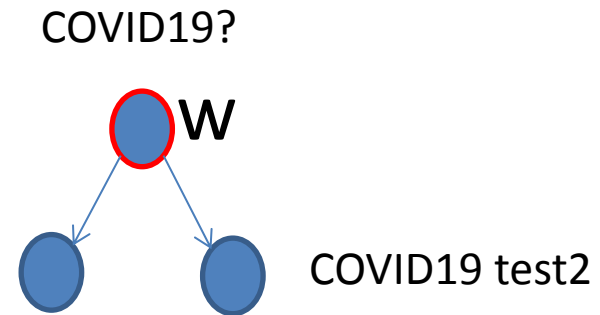


D-separation: Intuitive Examples

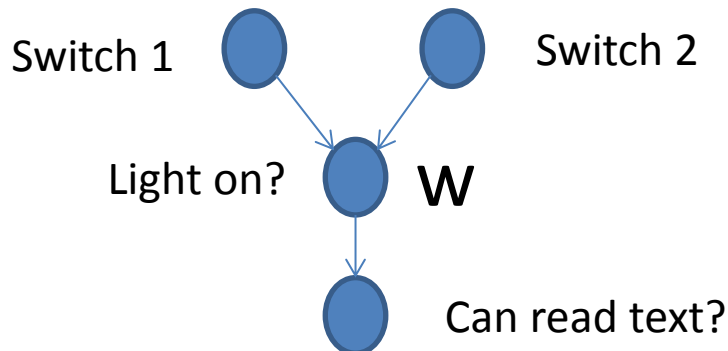
Passthrough



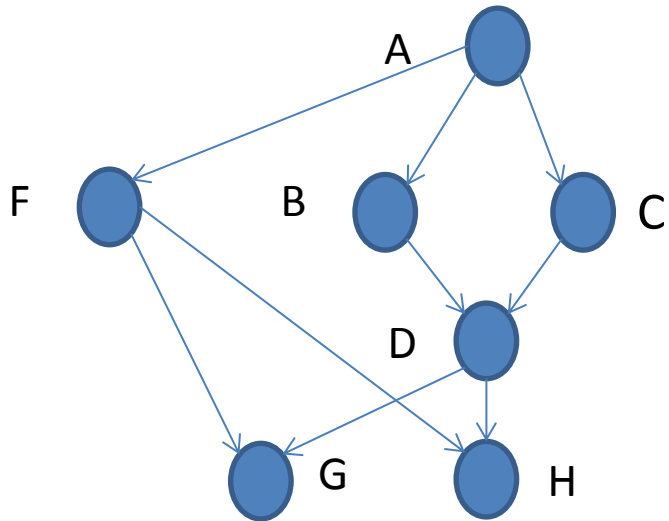
Diverging



Converging



D-separation: Exercise



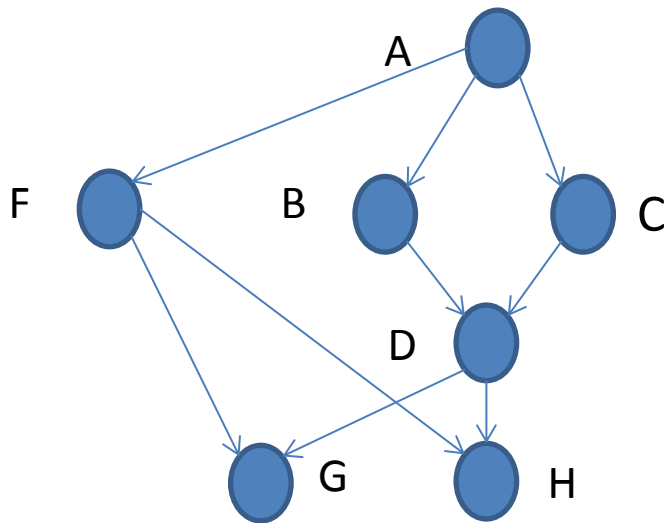
$d\text{-sep}(\{C\}, \{B, F\} | \{\})$? **NO (B-A-C) unblocked**

$d\text{-sep}(\{C\}, \{B, F\} | \{A\})$? **YES**

$d\text{-sep}(\{C\}, \{B, F\} | \{A, D\})$? **NO**

$d\text{-sep}(\{C\}, \{G\} | \{D\})$? **NO**

D-separation: Efficient Scheme



$d\text{-sep}(\{C\}, \{F\} | \{A\})$?

Barren node: childless node not in X, Y, Z

Remove H, G (barren nodes)

Remove D (now barren), and then B.

Can now remove A (root evidence node)

No paths remain... so d-separated.

Bayes Network Structure Learning

Two main paradigms:

- 1) Statistical learning, usually Bayesian
- 2) Statistical independence tests.

Each paradigm has numerous models and algorithms. In general problem is NP-hard.

Statistical BN Structure Learning

Need to compute $P(h \mid \text{data})$ for some BN hypothesis including **structure** and **parameters**.

In some cases also have interesting $P(h)$, usually favoring simpler BNs (fewer arcs, smaller CPTs).

Statistical BN Structure Learning

Simplest example is MDL, where $P(h)$ is encoding length for network.

Encoding arcs for nodes: $L1 = \sum(k_i * \log(n))$ ($n = \#vars$)

Encoding CPTs: $L2 = \sum(CPT_i \text{ size} * \log(N))$ ($N = \#examp.$)

Minimize encoding size, including length of describing data given the hypothesis.

That is, minimize:

$$DL = L1 + L2 - \log(P(\text{data} | \text{hypothesis}))$$

These can be evaluated “locally”.

Numerous algorithms to search for “best” structure.

Independence-Test Schemes

Preform various statistical tests:

$\text{Independent}(x,y|Z)$? (Based on the data)

E.g. $\text{Independent}(A,B|\{ \})$?

$N_a/N * N_b/N \approx N_{ab}/N$? (etc.)

If test passes, decide $d\text{-sep}(x,y|Z)$

and construct network accordingly.

Numerous schemes determine how to select independence tests to do, and how to construct the network efficiently.